

HEALTHCARE DATA INITIAL ANALYSIS REPORT

1. Purpose of this document

This document records my understanding of the HealthConnect no-show problem, my review of the project resources, my first read of the appointment dataset, the business questions I will pursue, the KPIs I propose, and my analysis plan. It also lists the assumptions, limitations, risks and dependencies that could affect the work. As the Week 4 brief requires, KPIs here are identified and justified only; calculation and dashboarding begin in Week 5.

2. Understanding of the business problem

HealthConnect Clinic runs appointment-based services and loses capacity when patients do not turn up. The brief describes the pains: missed appointments, wasted slots, repetitive patient enquiries, and no clear picture of why no-shows happen. The central project question is how data and AI can reduce missed appointments and improve patient support.

My track (Data Analytics) owns the "understanding" half of that question: measure the size of the no-show problem, describe where and when it concentrates, and identify the factors associated with it (reminders, lead time, patient history, distance, scheduling). The other tracks build on this work, the Machine Learning track will predict no-shows and the Generative AI track will handle patient enquiries, so my output must be a clean, well-documented evidence base they can reuse.

Why this matters to the clinic: every no-show is a slot that could have gone to another patient, plus staff time lost. If we can identify which appointments carry high risk (for example long lead time plus previous no-shows plus long distance), the clinic can target confirmation calls and reminders where they matter most instead of reminding everyone.

3. Resources reviewed and how I used them

- **HealthConnect_Appointment_Data.csv** — 5,000 appointment records, 18 columns. My primary analysis input; structure and quality checks in Sections 4–5.
- **HealthConnect_Data_Dictionary** — definitions, types and notes per variable. It guided every quality check, e.g. the rules "previous_no_shows cannot exceed historical appointments" and "distance and waiting time contain limited missing values," both of which I verified in the file.

- **HealthConnect_Clinic_Knowledge_Base** — clinic procedures including rescheduling and cancellation rules. I reviewed it to understand how a cancellation differs from a no-show, so I treat them separately in all rate calculations. (It is primarily a Generative AI track resource; I used only the procedure definitions.)
- **Week 4 assignment brief** — tasks, outputs and criteria. It shaped this document's structure; Appendix A maps each criterion to a section.

Discrepancy found: the dictionary describes the dates as ISO format, but the delivered CSV uses M/D/YYYY (e.g. 2/6/2025). I logged this as a Week 5 cleaning task (Section 7).

4. Dataset overview

- **Size:** 5,000 rows (appointment_id HC-00001 to HC-05000), 18 columns.
- **Grain:** one row per appointment. patient_id repeats across rows, so patients appear multiple times, this allows patient-level history analysis.
- **Period:** bookings observed from around Nov 2024 to Jun 2026; appointments from around Jan 2025 to Jun 2026 (roughly 18 months).
- **Target variable:** appointment_outcome, with three values: Attended, No-Show, Cancelled.
- **Variable groups and analytical role:**
 - Patient demographics: gender (Female/Male/Prefer not to say), age (18–80, adults only), age_group (six bands), describe *who* misses.
 - Scheduling: appointment_type (4 categories: General Consultation, Specialist Consultation, Follow-up, Diagnostic Test), appointment_day (all 7 days, the clinic operates daily), appointment_time (Morning/Afternoon/Evening), booking_lead_days (0–60 observed), describe *when and how* appointments are booked.
 - History: previous_appointments, previous_no_shows — behavioural risk.
 - Reminders: reminder_sent (Yes/No), reminder_channel (SMS/WhatsApp/Email; "None" when no reminder), the clinic's main intervention.

- Access and experience: distance_to_clinic_km (0.5–45 observed), waiting_time_minutes (2–68 observed).

5. Initial data quality assessment

Completeness

- distance_to_clinic_km has limited blanks (e.g. HC-00039, HC-00177), matching the dictionary note.
- waiting_time_minutes has limited blanks (e.g. HC-00055, HC-00162), same note.
- All other columns appeared complete in the rows inspected.

Consistency

- reminder_channel is "None" on every row where reminder_sent = No, and a real channel where Yes. This is a designed dependency, not an error; I will not treat "None" as missing.
- previous_no_shows never exceeded previous_appointments in the rows inspected (e.g. HC-02006: 9 previous appointments, 3 previous no-shows), matching the dictionary rule.
- age and age_group agree on spot checks (e.g. age 78 in band 65+).

Validity and ranges

- age 18–80, consistent with "adult patients only."
- booking_lead_days 0–60; 0 means same-day booking and appointment.
- distance 0.5–45 km; waiting 2–68 minutes. A few waiting values above 50 minutes are high but plausible; I will flag them as outliers to review, not delete.
- appointment_type, appointment_time, appointment_day and appointment_outcome contain only their expected categories.

Uniqueness

- appointment_id is sequential and unique (primary key). patient_id is intentionally non-unique.

Initial outcome mix (indicative only) A manual scan of the first 100 records showed roughly half Attended, about 4 in 10 No-Show, and a small share Cancelled. This is a rough sense-check, not a final figure — exact rates will be computed in Week 5. Even as a rough check, it confirms the no-show problem is large and real.

Handling decisions taken this week

- No imputation of missing distance/waiting now; related analyses will use non-missing rows.
- Cancelled is treated separately from No-Show in every rate (a cancellation frees the slot in advance; a no-show wastes it).
- The original CSV will not be altered; cleaning will live in Power Query as separate steps, per the brief's resource instructions.

6. Key variables for attendance and no-show analysis

- `appointment_outcome` — the target.
- `reminder_sent` / `reminder_channel` — the clinic's intervention; test whether reminders associate with attendance and which channel performs best.
- `booking_lead_days` — long lead times may mean forgotten appointments; informs when to confirm.
- `previous_no_shows` / `previous_appointments` — past behaviour is usually the strongest signal of future behaviour; informs targeted follow-up.
- `distance_to_clinic_km` — travel friction may drive no-shows; informs transport support or rescheduling.
- `appointment_time` / `appointment_day` — certain slots (e.g. Monday mornings or evening slots) may carry higher risk; informs slot planning.
- `appointment_type` — different services may behave differently.
- `age_group` / `gender` — descriptive context for communication targeting.
- `waiting_time_minutes` — experience factor; long waits may discourage attendance.

7. Business questions

Q1. What share of scheduled appointments ends in a no-show, and how does it vary by month, day of week and time of day?

Q2. Do reminded patients attend more, and which channel (SMS, WhatsApp, Email) has the lowest no-show rate?

Q3. Are patients with previous no-shows more likely to miss again?

Q4. Does a longer booking lead time associate with higher no-show likelihood?

Q5. Does distance to the clinic associate with no-shows?

Q6. Which appointment types carry the highest no-show risk?

8. Proposed KPIs (brief requires 3–5; I propose 5)

Note: per the brief, these are proposed and justified this week; they will be implemented as DAX measures in Week 5.

KPI 1 — Appointment No-Show Rate. No-shows $\div$ scheduled appointments (Cancelled excluded). Linked to Q1. Justification: the headline measure of the problem; tells the clinic how much capacity is wasted and gives a baseline to measure any intervention against.

KPI 2 — No-Show Rate by Reminder Status and Channel. The no-show rate computed separately for reminded vs not-reminded appointments and by SMS/WhatsApp/Email. Linked to Q2. Justification: reminders cost money and effort; the clinic must know whether they work and which channel to prioritise.

KPI 3 — Repeat No-Show Rate. The no-show rate among appointments of patients with at least one previous no-show. Linked to Q3. Justification: if a small group of repeat missers drives a large share of no-shows, the clinic can target them with confirmation calls instead of reminding everyone.

KPI 4 — Average Booking Lead Days (No-Show vs Attended). Average lead days computed separately for no-shows and attended appointments. Linked to Q4. Justification: if no-shows cluster at long lead times, the clinic can add a mid-window confirmation step for early bookings.

KPI 5 — No-Show Rate by Distance Band. No-show rate computed over bands (e.g. 0–5 km, 5–15 km, 15+ km). Linked to Q5. Justification: if far patients miss more, the clinic can offer rescheduling or transport guidance.

9. Initial analysis approach

Week 4 (completed): resources reviewed; structure and quality checks done; questions and KPIs defined; this document prepared. **Week 5 (planned):**

- Step 1 — Load the CSV into Power BI Desktop; in Power Query parse dates to true date type (fixing the M/D/YYYY vs ISO discrepancy), set numeric types, keep missing values as nulls, and add band columns (lead-time band, distance band). Original file untouched.
- Step 2 — Implement the five KPIs as DAX measures.
- Step 3 — Build descriptive visuals: outcome mix; no-show rate by reminder/channel, lead band, distance band, time/day and appointment type; monthly no-show trend.
- Step 4 — Cross-check key numbers with Excel pivot tables to verify the Power BI calculations.
- Step 5 — Record findings and hand clean measures to the ML and GenAI tracks. **Tools:** Power BI Desktop, Power Query, DAX, Excel for verification.

10. Assumptions, limitations, risks and dependencies

Assumptions: appointment_outcome is recorded reliably; Cancelled and No-Show are distinct behaviours and must be measured separately; the synthetic data behaves like a real clinic's appointment data for this exercise.

Limitations: no field explains *why* an appointment was missed or cancelled; no external factors (weather, transport, illness); waiting_time_minutes is an estimate per the dictionary; single clinic, so findings may not generalise.

Risks: missing distance/waiting values could bias the access analysis if not missing at random; small subgroups (e.g. gender "Prefer not to say") are too small for reliable comparison; if the date-format issue is not fixed before time analysis, monthly trends will be wrong; correlations here do not prove causation, so recommendations must be worded carefully. **Dependencies:** the data

dictionary for definitions and the knowledge base for cancellation/rescheduling rules; Power BI Desktop access; alignment with the ML track (same cleaned data feeds the prediction model) and the Project Management track's timeline.